The Agency of Responsive Envelopes: Interaction, Politics and Interconnected Systems

Geoffrey Thün, Kathy Velikov, Mary O'Malley and Lisa Sauvé

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Abstract

This article positions the territory of responsive envelopes within the context of contemporary disciplinary questions surrounding the politics of the architectural envelope on one hand, and the agency of material explication of environmental, social and spatial performance on the other. Two recent prototype-based responsive envelope projects undertaken by the authors, the *Stratus Project* and *Resonant Chamber*, are described in detail relative to the reciprocity between the development of their materiality, form, production methods and their dynamic interaction with external forces, environments and inhabitants. An argument is made that responsive envelopes, in their capacity to structure continually evolving energetic, material and information exchanges between humans, buildings and the wider environment, have the potential to actively construct and enable political participation through spatial transformation, data driven processing and informatics. These envelopes are positioned as agents within wider ecologies and social systems, and as sites for the design of robotic architectures to engage such questions.

1. INTRODUCTION

> "Space is not a representation of a politics that would already otherwise exist in the abstract. Politics operates and flows through and in spatial practice." [1]

In his 2008 essay, "The Politics of the Envelope", Alejandro Zaera Polo positions the building envelope as the site of architecture ultimately loaded with political content, the "effective link between material organizations and politics...central to a political discussion of material practices" [2]. The envelope is politically loaded through its multifarious attachments to the politics of construction and representation: be it through its cultural attachments of image and form; or through its real engagement with material and energetic flows, environmental performance, labor relations and nonhuman actors; or through its effective, psychological and intellectual impacts. The building envelope thus emerges as a contested and deeply entangled disciplinary concern that warrants a renewed investment within disciplinary discourse. Although he frames the argument through the physical-formal taxonomies of the building envelope, it is particularly interesting to note that Zaera Polo identifies the primary political performance of both "flat horizontal" and "spherical" envelopes to be the development of a "political ecology" through their characteristics of management of environmental flow and exchange, and goes on to add that this "engagement with ecological concerns is contemporary architecture's most direct path to political effect" [2]. Within this framework, the responsive envelope might gain specific agency in its potential to actively construct and materialize a continually evolving politics of environmental and social participation.

Responsiveness is most simply used to describe, "how natural and artificial systems can interact and adapt" [3]. In architecture, it is a term that can refer to both machine led-systems control and user-led control [4] and has been used interchangeably with the terms "adaptive" and "interactive" [5, 6]. For the purposes of this article we propose to define a responsive envelope as a material assembly constituting a spatial boundary that combines kinetic components with computationally driven sensing and actuation regimes that dynamically adapt to both environmental and human interaction within a learning system [7]. Building envelopes are increasingly being developed as complex systems of material and technological assemblies and can often be considered quite literally "skins", approaching biotic membranes capable of sophisticated energetic, material and information exchanges [8]. Although the discipline has seen a growing number of buildings and installations wherein environmental information is enabled through the communicative interfaces of "aware" building elements [6], the agency of this information and its potential to foster forms of micropolitics and human action is still an open question [9]. In order for these works to gain social and political agency, it appears that an area of

further exploration needs to take place in the territory of the nature of human interaction and *action(s)* that these works might produce. While notable pioneering installations include the work of Philip Beesley [10], Usman Haque [11], Rafael Lozano Hemmer [12], and Omar Khan [13], this remains a potent area of architectural design exploration and experimentation: one where an expansion of instrumental system performativity and more complex coevolution with human agents is anticipated. The responsive envelope might constitute a ground that works to simultaneously enable both the symbolic order (social and political order of humans) and the material world (the sciences), thereby undermining the false ontological and epistemological duality that Latour identifies Western culture has operated within since the Enlightenment [14].

The work discussed in this article focuses on two responsive interior envelope prototypes that pursue these questions: the *Stratus Project* and *Resonant Chamber*. Both projects aim to develop active, distributed envelope systems that operate through continual information and material exchange, and eventually dialogue, between humans and the soft systems of architecture, such as light, thermal gradients, air quality and acoustics. There has been a surfeit in recent years of research and development of advanced exterior building skins, often with intelligent and kinetic features focused on environmental performance. Our attention is turned to the interior and how the thickened, responsive envelope might begin to condition new experiences of interiority and envelopment that move beyond optimization? Embedded sensing, active control, mechanical actuation and information processing are mobilized within these projects to enable architectural interfaces that can become media of communication and mutual relation, while also "structuring participation" [9] - both spatial and social. We elaborate on these projects not only through their performative and material effects, but also through a detailed account of the process driven research in digital design, simulation, manufacture and testbed installation undertaken in their ongoing development.

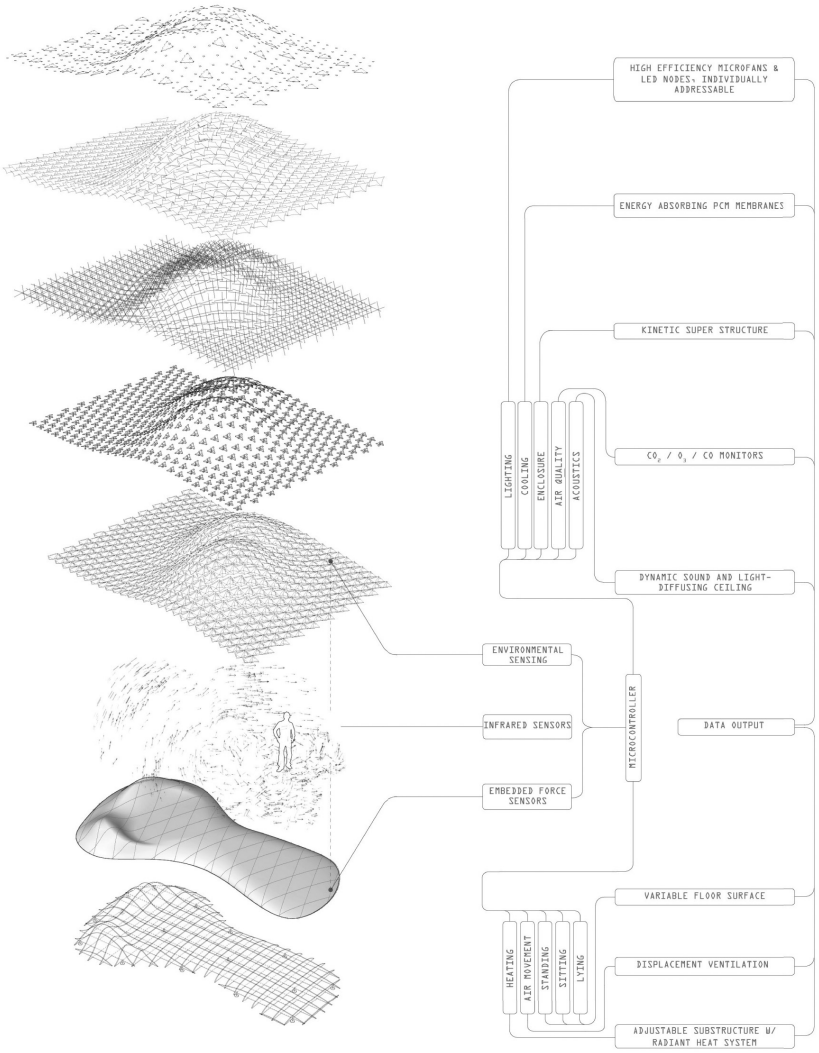
2. THE STRATUS PROJECT

The Stratus Project "Figure 1" involves research and prototype development towards a personalized and distributed, kinetic light and air-based interior envelope system. The system is comprised of a deeply arrayed suspended textile - a 'thick' surface or epidermis, consisting of a networked matrix of physical elements and technologies that sense and modify space and environment based on human presence, temperature, humidity and airborne pollutants [15]. Within the domain of envelope system development, the project explores the physical, aesthetic and operational possibilities of building envelopes conceived of though the principles of complex systems [16], that entail distributed, discrete and redundant elements interacting with and producing environments through data-driven performance.

► Figure 1: The Stratus Project v1.0 installed, January 2011



Formally, the project builds on the possibilities of kinetic, tessellated surfaces to achieve dynamic architectural environments based on variable inputs, initially explored by Mark Goulthorpe and his collaborators in the "Aegis Hyposurface" from 1999-2001, and on a number of projects exploring emergent adaptive surfaces such as Rob Ley and Joshua Stein's 1999 "Reef", Marilena Skavara's 2009 "Adaptive Fa[CA]de", and the adaptive facade systems being developed by Chuck Hoberman and Buro Happold within their collaborative ABI project. The Stratus Project engages the air environment as a primary sphere of biopolitcs that has emerged at the forefront of contemporary philosophical and disciplinary discourse [17, 18, 19, 20]. Through ambient information cues that parallel technical performance, this sensing, physical envelope is intended to operate as an interface through which inhabitants might develop not only more sensible and cognitive relationships between their own actions, the spaces they inhabit and the larger air environment, but also with the technologies and political apparatus [21] that exercise control and agency in the "spheres" of our existence [22]. Stratus v1.0 was constructed as a 2.4m x 4.0m working prototype installation aimed at testing the physical components, human interface and the frictions and conflicts in preliminary controls logics. The system forms a dynamic enclosure intended to be paired with a piezoelectric activated displacement ventilation system volumetrically and materially adjusting the environment via a complex of sensing and actuation regimes "Figure 2".



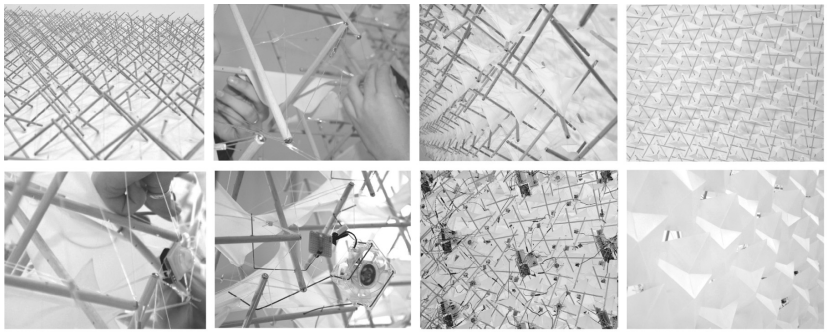
◀ Figure 2: Exploded axonometric drawing of layered physical components and logic streams of Stratus controls.

2.1. Structure

Exploration into various types of kinetic structures ultimately settled on a tensegrity-based structure that was selected for its lightweight and stable properties, as well as its performance within a deformable, kinetic system [23]. The cable strut configuration used for the prototype was adapted from a system developed by mathematicians Wang & Liu [24], with each unit comprised of three rods connected with elastomeric cord. This structural base provides controlled deformational flexibility, while also supporting a network of sensors, actuators, lights, micro-fans, Arduino microcontrollers, and light diffusing fabric panels impregnated with phase-change coatings for

thermal performance "Figure 3". Attached to the underside of the structure are the *breathing cells*: a translucent, light-diffusing skin, locally actuated for air movement "Figure 4, right".

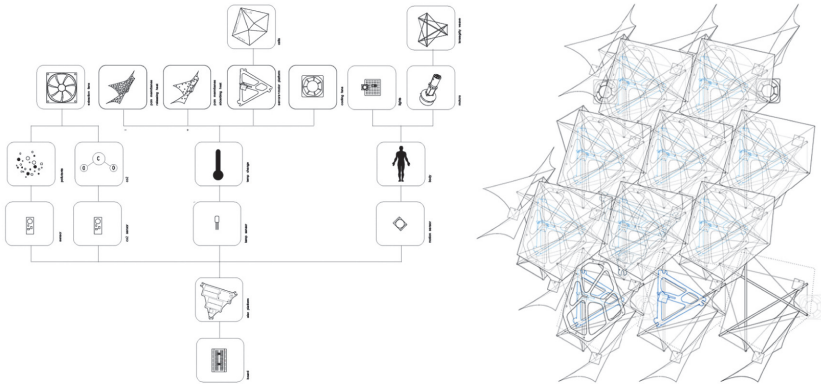
► Figure 3: Assembly of components of Stratus prototype v1.0.



2.2. Circuit and Sensing

As Stratus v1.0 was a demonstration prototype, simple, inexpensive, off-the-shelf electronic components, currently common in interaction design, were used. A circuit consisting of a microprocessor, servos, LED lights, sensors and micro-fans forms the basic operational building block of the complete installation "Figure 3, left". Each circuit module mounts to the top of the tensegrity weave and controls a territory of six cells. Within each circuit, an Arduino Uno prototyping board reads input from one temperature sensor and one passive infrared (PIR) motion sensor, and controls output to the micro servos, LED lighting and the mini-fan. Temperature sensors are located on each electronics platform and passive infrared (PIR) motion sensors are distributed in a tight grid within the system to target fine-grained zones of thermal change and human activity.

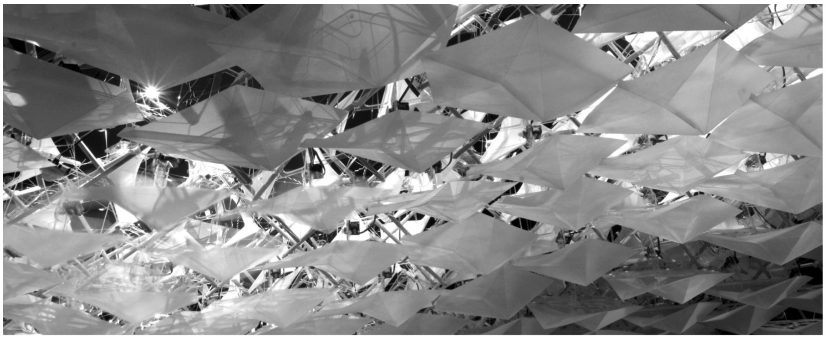
► Figure 4: (left) Stratus Project v1.0 circuit logic and components; (right) plan of system array.



The control program is composed in the Arduino programming language and is written to allow for situational adaptation both upon initial installation and while in operation. Once the Stratus Project is operational, the program initiates baseline ambient readings to set thresholds and then calls on a number of subset programs to control the separate systems of lighting, cooling, and cell activation. Custom Boolean logics generate feedback loops that inform the Stratus Project when it is in an active cooling state or a passive state, and a similar logic is used to track light intensity levels in relation to occupancy. Tracking the operation of the servo actuators mitigates electrical interference that could disrupt the analog sensors.

2.3. Breathing Cells

Breathing cells "Figure 5" comprise the primary visible surface of the Stratus Project. Each cell is made from a single die-cut sheet of translucent rigid polymer, folded and adhered into form, and flexibly attached to the substructure. These cells are designed like scales, overlapping to provide a tight continuous surface across multiple formations of the ceiling plane when closed. Motion is triggered by input from the temperature sensor so that when the temperature threshold is exceeded, micro servo-motors open and close localized groups of cells to permit ventilation and airflow. A Variable Speed Servo library [25] is used to choreograph the movements of the cells within each circuit, minimizing conflict. Zones of lighting respond to occupancy movement and activity through location tracking and brightness. We are currently developing genetic computational algorithms which will allow the system to self-adjust and learn over time, both relative to variable environmental conditions and to emergent occupancy patterns, preferences and habits, via historical data logging. Our ambition is that this envelope will begin to exhibit "intelligent" characteristics associated with second-order cybernetics.



◀ Figure 5: Installed Stratus Project v1.0 prototype: breathing cell array.

2.4. Air Quality

In addition to the circuit modules that control lighting, cooling, and cell movement, two additional circuits control an array of fans for air extraction. These circuits consist of an Arduino microcontroller, a CO_2 sensor, and extraction fans. In this application, the CO_2 sensor has been outfitted with a prefabricated circuit that allows one to manually set a threshold level for CO_2 concentration. An exceeded threshold level triggers the extraction fans to displace stale air and permit the movement of fresh air into the space and can be coordinated with magnetic dampers in the displacement ventilation system at floor level. Blue LED lights indicate when air extraction fans are in operation, complimenting sensible levels of air movement with ambient communication by alerting the breather of the condition of reduced air quality and their possible agency within that condition. "Figure 6" Through these ambient cues, the physical envelope will begin to operate as an interface through which inhabitants might develop more sensible and cognitive relationships between their own actions, the spaces they inhabit, and the larger air environment.



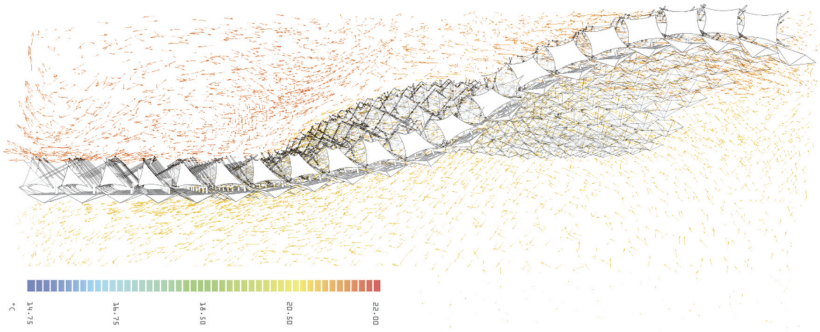
► Figure 6: Breathing cell field closed, in operation, and with air extraction fan deployed.

Running concurrent to the operation of the air environment modifying apparatus, a custom graphing program provides visual feedback on atmospheric conditions. This program also employs an Arduino platform to intake data readings, which interfaces with a Processing-based animation displayed on an adjacent wall panel. This communicates CO_2, VOC, and humidity through relative rather than quantitative graphics, tracking variation over time rather than specific parts per million levels or percentage humidity, with the intent of this information being accessible to non-expert inhabitants. Eventually, this feedback could be available through a smart phone device or linked to individual preferences registered through RFID signatures for frequent inhabitants.

2.5. Thermal Response

Working in conjunction with the breathing cells and occupancy detection, localized heat relief is provided by distributed micro-fans mounted within

the tensegrity weave. When the ambient temperature exceeds the established threshold and the cells open, the fans activate to provide localized sensible cooling. Further thermal modification occurs through a field of tensioned fabric panels located above the lighting array and impregnated with phase change materials (PCMs), a family of smart materials that can absorb or release energy while the material changes state, and so can contribute to the thermal stabilization of interior spaces. The fabric layer is designed to maximize its exposed surface area in order to provide as much absorptive PCM area as possible. At a larger scale, gross movement of the entire system is able to create a stratification of thermal gradients "Figure 7", reducing the amount of air to be conditioned when a space is unoccupied and limiting energy devoted to inactive spaces.

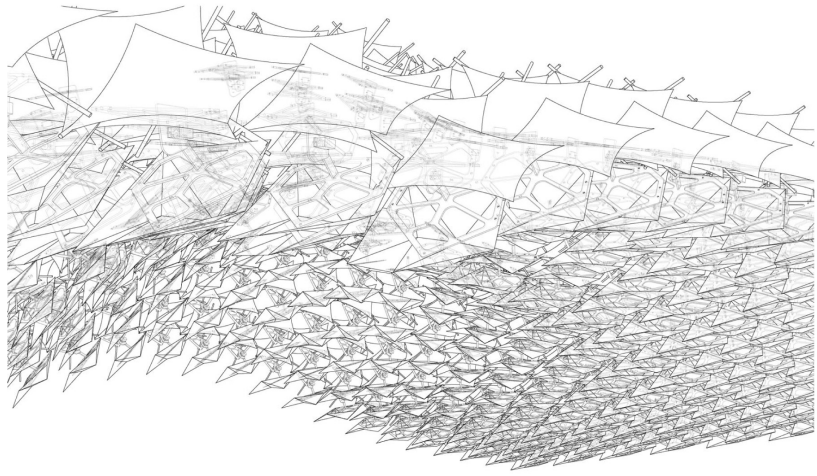


◀ Figure 7: Fluent® model simulation indicating thermal stratification relative to spatial and cell configuration.

2.6. Dynamic Spatial Transformation

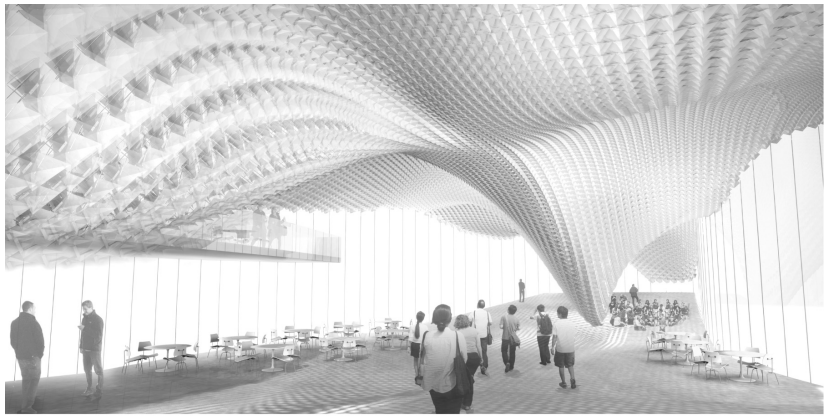
The tensegrity structure and the overlapping tessellation of the cells were designed to accommodate dynamic movement to achieve variable spatial configuration of the surface "Figure 8". The prototype is outfitted with six stepper motors, each with a dedicated microcontroller, controlled through a single Arduino. The actuated gross displacement of the envelope uses feedback from occupancy and position sensing to inform modification of its overall spatial volume, allowing the variable creation of zones of intimate or collective scale. Through the gross deformation of this envelope, the volume of conditioned space is altered in response to occupation. This, combined with occupant-based localized ventilation, is anticipated to minimize the energy consumption associated with air delivery, as compared with conventional systems [26] while providing a much more interactive and fine-grained approach to air quality provision driven by inhabitation.

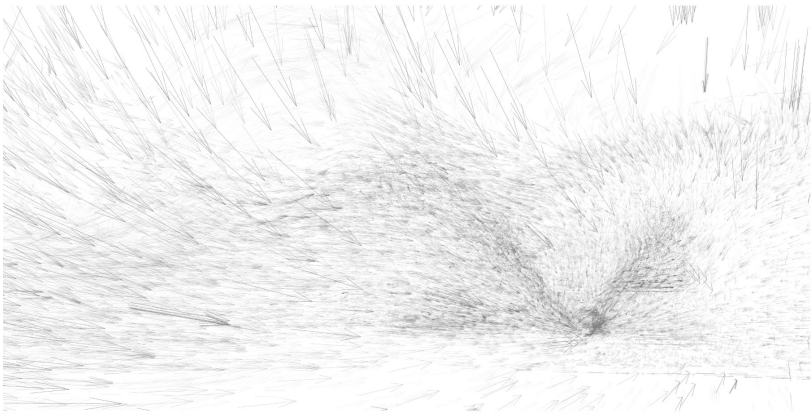
► Figure 8: Stratus Project array deformation shapes the volume of conditioned air in response to occupant presence and demand.



The Stratus Project thereby has the ability to produce flexible architectures where both microclimate and enclosure co-define and co-condition activity and use "Figure 8". The project works to reclaim the environmentally performative domains of architecture – in this case, specifically, interior mechanical delivery and interface systems – to within the purview of the discipline, as territories of material, formal, technological and experiential innovation and exploration. The intention of the first prototype installation of the Stratus Project was to explore a complex, multifunctional, multi-agent system for interior envelopes. Beyond the specific technological refinement of the system itself, it is intended that the Stratus Project exploration be expanded to the full scale of a building environment, where it might become a fully integrated interface and responsive interior envelope that, through its ability to be locally tuned and communicative of its air quality status, might literally become an extension of individual and collective bodies and their atmospheric spheres "Figures 9, 10".

► Figure 9: Illustration of internal view with articulated variable microclimate and spatial zoning provided by the Stratus system.





◀ Figure 10: Air flow velocity distribution diagram produced through FLUENT© complex fluid dynamics simulator.

3. RESONANT CHAMBER

In his seminal text "Homo Ludens: A Study of the Play-Element of Culture", Johan Hunziga describes how the play-form is a practice through which the fundamental forms of culture and social life arise - from law and war to knowledge, poetry, art and philosophy [27]. Cybernetic interaction, in many ways can be understood as the logic system that defines play-forms as processes of developing social intercourse and learning [7], hence not only its propensity in gaming environments but also why game theory can often be used as a basis by which to model complex adaptive systems [16]. While Hunziga, a consummate humanist, may not have imagined his theory to extend to nonhuman entities, this suggests implications for responsive envelope design that may expand this notion to develop new types of exchanges and practices between people and built environments. Within the history of aural space, there is speculation that musical and performative forms have coevolved concurrently with the formal and material qualities of architectural space in which they were practiced: for example, one can look to the evolution of medieval plainsong in the highly reverberant cathedrals to the development of elaborate counterpoint in the less lively churches of the German reformation [28], or musical works designed for the highly specific acoustics of particular buildings [29].

Resonant Chamber "Figure 11" continues our exploration into interior responsive envelope systems, now turning to the possibility of transforming the acoustic environment through dynamic spatial, material and electro-acoustic technologies. It is intended to create an *instrument* at the scale of architecture, capable not only of dynamically adjusting spatial properties in real-time response to changing sonic conditions but also inviting new forms of play, performance and interaction to develop. The system is designed as an enveloping soffit consisting of a geometrically segmented surface that can be altered through localized and gross deformation to modify both surficial

characteristics and overall spatial volume to transform its acoustic performance. Unlike the approach deployed in the Stratus Project, Resonant Chamber is designed as an infrastructure for responsive actuation whose constituent components and operational logistics are developed independent of a range of possible sensing and actuation protocols. This characteristic leaves it open for variable development with respect to optimization and interaction regimes.



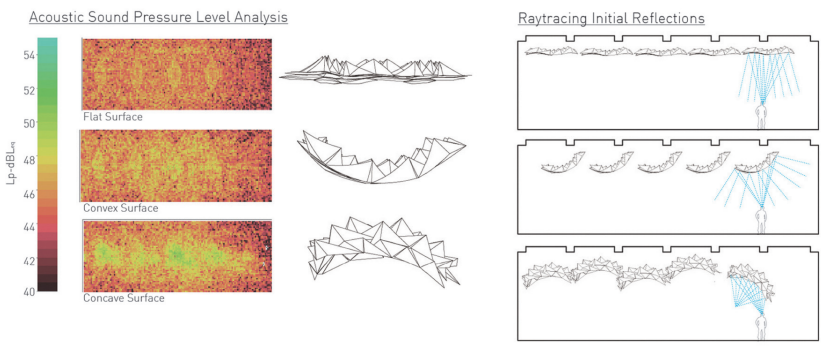
► Figure 11: Resonant Chamber: (left) prototype of three clouds, installed January 2012; (right) underside of performative acoustic surface of absorptive, reflective and sound generating panels.

Physically, this work shares a lineage of development with kinetic tessellated architectural systems and variable acoustic surfaces that includes David Serero's 2005 "Variable Geometry Acoustical Domes", the current research of Brady Peters, particularly his 2011 "Distortion II" and Eddy Sykes' explorations in kinetic rigid origami structures. Resonant Chamber advances this line of inquiry by exploring the application of multi-functional 'thick' material treatments within a volumetrically variable acoustic space paired with kinetic operation digitally controlled via environmental sensing and self-regulating systems.

3.1. Structure and Geometry

Whereas the Stratus Project explores tensegrity as the basis for an integrated, deformable structural system, the Resonant Chamber investigates rigid origami as a flexible geometric system that makes it possible to achieve predictable formal outcomes through the variation of a continuous surficial assembly. Rigid origami's surface properties of developability, flat-foldability (folded from a single sheet) and controlled

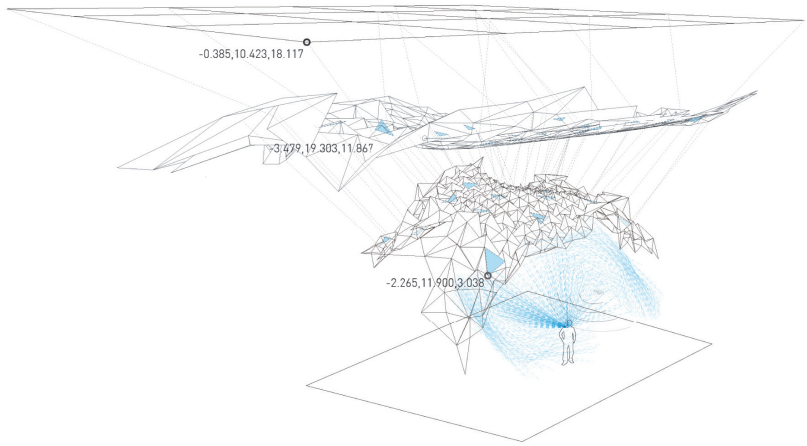
transformation, make it possible to substitute its facets and fold lines with rigid panels and hinges, and enable its application for kinetic structures [30]. As a result of the interconnected reactions between interior vertices and crease lines and by virtue of the linked structure, physical actuation in one location has calculable effects on the movement of adjacent elements, thus reducing the number of points of actuation required to induce overall transformations. This geometry allows for gross deformation of the surface to dynamically alter spatial volume to produce changes in reverberation time and also control early acoustic energy sound waves and reflections "Figure 12". It also allows for localized surface folding to vary the ratios and orientation of exposed surfaces with different material and acoustic properties for late acoustic energy control. Although there are precedents in control of late room response through systems that control height or orientation of a ceiling reflector [31], Resonant Chamber develops a kinetic system that can simultaneously control wave front curvature, level, and time of arrival of early reflections.



◀ Figure 12: CATT acoustic sound pressure and ray-tracing analyses of geometries produced by ARUP Acoustics.

Real-time responsive digital models were used to anticipate the geometric limits of the dynamic surface and to provide information for the geometric constraints and actuation logics that would transform the surface in space. Basic origami software Freeform Origami developed by Tomohiro Tachi provided fundamental origami meshes, which were then further developed through Rhinoceros 4.0 software. The Rhinoceros plug-in Grasshopper with Kangaroo enabled live physics simulation to predict the dynamic transformation while accounting for gravity and linkage forces "Figure 13".

► Figure 13. Modeling of spatial transformation of the system illustrating the relative displacement of a single point tracked between gross and surficial deformations.



Relative displacements between panels were extracted from the Rhinoceros model and distributed as length outputs via Firefly to an onboard Arduino to control the folding of the cells "Figure 14". Two modes of actuation are required within the system to control various types of acoustic response: linear actuators mounted to panels alter material exposure through localized deformation of absorption cells while stepper motors suspend the system to adjust gross displacement of the surface.

► Figure 14. Sensing, processing and linear actuator system, seen from above, controls surface deformation and folding (left); folded surface seen from below (right).

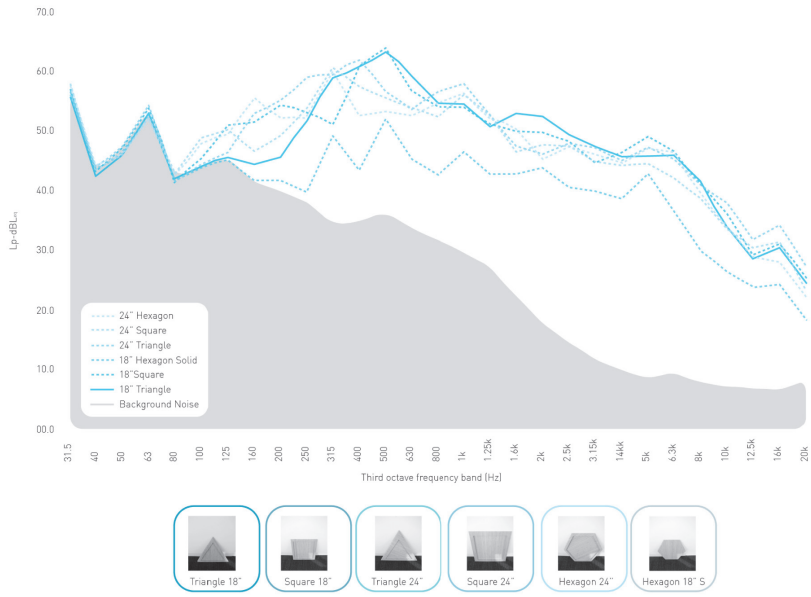




3.2. Physical Prototype Development

Acoustic performance depends both on the geometry of spatial enclosure as well as on specific material characteristics of bounding surfaces. While simulation software can be used to accurately predict the acoustic performance of architectural space, confirming the material characteristics of specific assemblies and geometries requires physical prototyping and verification in order to inform the assumptions in the model [32]. Developed in collaboration with ARUP Acoustics, potential material assemblies for Resonant Chamber's reflective and absorptive panels were

evaluated through acoustic simulations as well as 1:1 scale physical prototype testing to determine optimal panel geometry and material characteristics relative to their acoustic performance "Figure 15".



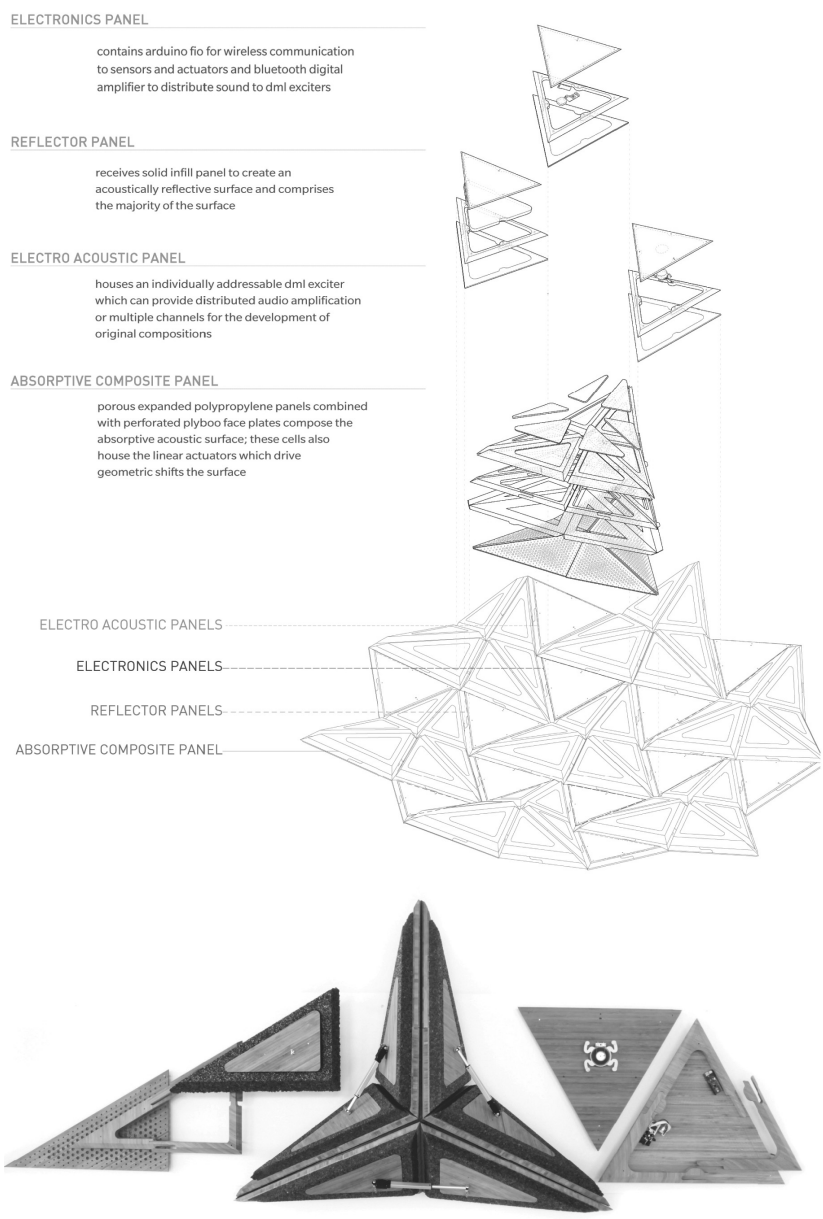
◀ Figure 15. Acoustic sensitivity comparison of panel shapes in bamboo plywood for identical DML exciter and amplification settings. The 18" triangle, square, and the 24" triangle performed within the ideal dB range of the frequency band. The 18" triangle was ultimately chosen for the prototype.

Through the evaluation of the formal capabilities of particular origami patterns and physical testing of acoustic and electro-acoustic performance, it was determined that an 18" plywood triangle would allow for optimum performance. Based on this, the team decided to develop a full-scale system prototype using a tessellated pattern first developed by Ron Resch [33], which is comprised of two sizes of triangular cells.

The introduction of distributed mode loudspeaker (DML) exciters embedded within the system, that effectively turn select panels into a system of distributed speakers, added further complexity to both predictive simulation and performance. Integration of electro-acoustics provides for an augmented level of reverberation control as well as directional sound reinforcement, which can then be actively manipulated for greater acoustic control. While electro-acoustic controls have previously been combined with static acoustic surfaces [34], integration within a kinetic surface has not yet been explored. This material-electronic system now avails an entirely different interactive interface from the spatial-material sound control approach of the physical system, opening up a variety of possible applications for interactive sound installations, immersive live performance spaces or acoustically enhanced learning facilities.

Three principal types of performative composite panels - reflective, absorptive and sound generating (via electro acoustics) - were developed within a standardized laminated panel framework, "Figure 16". The basic panel framework allows for the possibility of systemic variability within the composition of the overall system through the interchangeability of panel insert and surface treatment. The acoustically reflective panels provide a 120 Pa (N/m^2) density, optimum for acoustic reflection. The absorptive panels use porous expanded polypropylene (PEPP) to provide the acoustic absorption coefficient desired for sound dampening effects. PEPP was selected for its ability to contribute to panel rigidity and for its ease of milling to achieve the desired geometric configurations for flat-folding origami. The PEPP panels are faced with a perforated surface allowing a minimum 25% exposure for optimal noise reduction with noise reduction coefficient of .85. In the prototype, these panels are located within the flat-folding cells whose surface exposure is altered through actuation, so that a fully contracted system exposes only reflective surfaces.

These acoustically performative composite materials required the development of specific geometries and flexible joints that allow for the flat folding of the thick surfaces [35]. Thus the translation of the nominally thick computational model into a physical artifact required feedbacks and iterations between geometric and material performance, as well as manufacturing logics (and limits) during prototype development. The underlying unit of joinery for the Resonant Chamber prototype is a flexural hinge between rigid panels, which operates analogous to the fold of the origami surface to which thick panels are mated. A membrane is mechanically fastened between individual panels resulting in a continuous surficial assembly that can vary in size producing small sound clouds of 10 m2 (as was demonstrated in the first prototype installation) up to large soffits for performance halls, up to 1000 m2 (for which surface actuation logics become far more complicated to design). The mechanical clamping of the flexible membrane also allows for the system to be customizable in configuration and extent, and demountable for temporary installation.



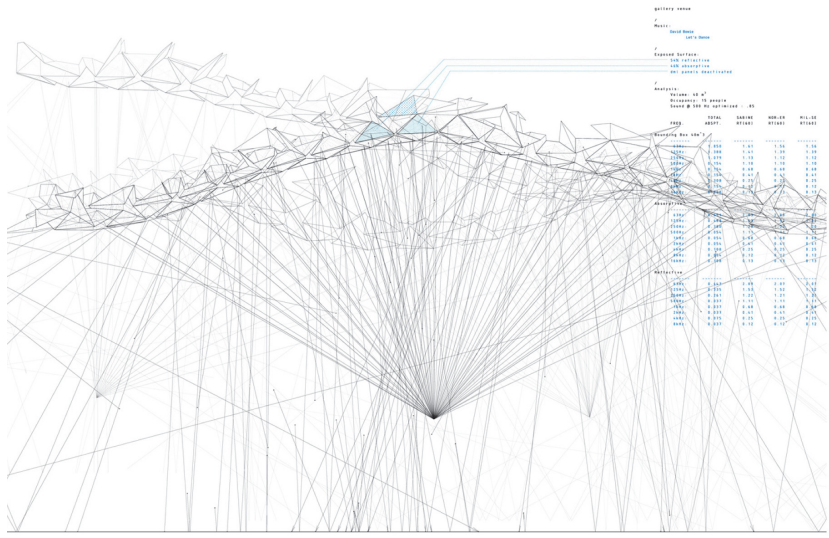
◀ Figure 16: Composite panel components, assembly and materials (above); prototype cell assembly (below)

3.3. Actuation and Communications

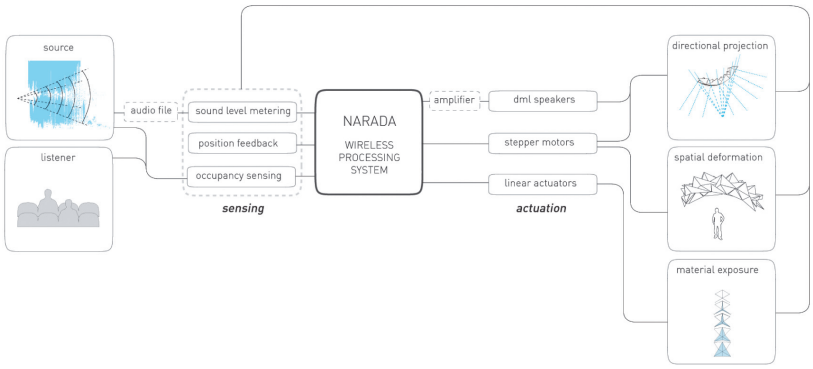
In order to enable actuation and communication, a fourth panel type within the system houses circuit and processing controls for linear actuation, sensing inputs such as local acoustic pressure and electro-acoustic amplification for the DMLs. The amplifier has multichannel capacity and can alter between a mono and stereo system to allow for a dynamic range of

► Figure 17: Acoustic reflections relative to envelope and spatial configurations.

electro-acoustic control. The first prototype of Resonant Chamber utilizes an Arduino fio, which receives wireless input from an xBee transmitter to control three sets of linear actuators. Coinciding with actuation of the surface are a series of cable-linked stepper motors which suspend the system and allow for overall geometric transformations, altering enclosed volume, panel position and interior surface curvature "Figure 17".



In the second prototype, currently in development, we are collaborating with Dr. Jerome Lynch at the University of Michigan to replace the Arduino system with his Narada-based system of communication. The Narada system achieves wireless sensing, actuation and data processing in a low power, high-resolution module. Its 16-bit analog-to-digital converter allows an array of sensors to be simultaneously interfaced with the device and locally processed to respond within the distributed system "Figure 18". Similarly, the 12-bit digital-to-analog converter can process voltage commands to the onboard potentiometer of the linear actuators to control stroke length and position feedback. On-board data processing offers distributed computational response rather than coordinating all inputs through a central programmed logic controller. This distributed processing system allows for custom, synchronized response across a continuous surface.



◀ Figure 18: Interrelated real-time sensing, acoustic control and actuation system logics.

3.4. Sensing Integration

The material systems infrastructure of Resonant Chamber is capable of being integrated with a range of sensing and actuation regimes. Although for the first prototype installation the system was set to operate through a pre-programmed loop, customized sensors and control logics can be embedded within the framework based on performance priorities regarding acoustic output and participant engagement desired by the designers and users. While pre-programmed modes are of course possible, our interest is in the dynamic and interactive capabilities of the system. Microphone-based systems can prioritize sound level samples to an average decibel range and inform the sound dampening cells to adjust accordingly, allowing more or less reverberation and optimizing aural conditions for listening to specific types of music or performance "Figure 19". Multifunctional sensors could dynamically track source and receiver and may be utilized to recalibrate performance spaces relative to audience position, number and spatial distribution. The system might also be embedded with learning capabilities, and priorities other than acoustic optimization, inviting new forms of interaction and live performance to develop. Alternatively, in a context of aural accessibility, inhabitant hearing needs can be isolated and customized for individuals through custom identifying receivers, tuning local aural conditions to the capacities of participants. These possibilities and their logics are being explored as part of the second generation of the project's development.

4. PROJECT(ING) RESPONSIVE ENVELOPES

When responsive architecture first entered design discourse in the late 1960's and early 1970's via figues such as William Brodey, Gordon Pask and Nicholas Negroponte, the conceptual high point entailed the development of a system that could operate through mutual dialogue with the environment *and* the inhabitant, and that could co-evolve through

► Figure 19: Illustration of a potential installation of Resonant Chamber system for a live musical performance.

underspecified means [36]. This notion of responsiveness challenges both the largely symbolic embodiment of politics as indicated by Zaera Polo, and the purely functional or managerial goals of most performance driven architecture. In our work within these types of systems through the Stratus Project and Resonant Chamber, we aim toward the development of an ecology of functions, performative and relational elements, that thicken the potential for robotic architecture by expanding its goals beyond environmental optimization, or even environmental awareness, towards an expanded idea of human agency and relations. The systems are often comprised of multiple, simple, off-the-shelf electronic components combined with custom fabricated material structures. Elements are often redundant and not necessarily perfectly optimized toward single functionality but evaluated for fitness toward multiple goals. Functionality becomes a "middle-out" process, emerging from a combination of bottom-up and controlling principles [37].



In the case of the Stratus Project, the full integration of sensing apparatus, distributed kinetic surface structures and air-modifying components delivers a system that simultaneously attends to issues of thermal control and individualized occupant comfort while rendering legible through haptic cues the more troubling questions that surround matters of air quality and the agency of inhabitants within a context of gaseous chemical dependency. In the case of Resonant Chamber, the responsive envelope is conceived of as a dynamic surface of composite material configuration that through volumetric and surficial deformation offers a model of responsive architecture whose performance can be altered to construct a range of

possibilities, enabling equally optimization of dynamic aural adjustment, the reconfiguring of relations between audience and performer, or an architectural-acoustic interface capable of engaging with inhabitants through forms of interactive play.

The responsive envelope offers an architecture that is no longer stable and predictable but negotiated and contingent, actively engaging in human participation within environmental and social processes, insistently public in its material and informational explication of these processes. Perhaps most importantly, the responsive envelope might even be able to challenge values and expectations regarding the role of architecture, and suggest a different ontological relationship between humans and things - one far more based on mutual exchange and complex entanglements and interrelationships of which we are seldom in control.

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